

Fact Checking Scribbles

- Singer's theory about Tuesdays are correct as they do have lowest energy scores.
- Singer's hypothesis about Thomas is wrong, as when volume level increases and crowd decreases the energy do not decreases and the venue The Vampire's Den must have a volume limit as they have lowest average volumes as compared to other venues.
- The Vampire's Den definitely becomes more lively at night it is not true for all venues as monks prefer morning shows and snobs prefer afternoon shows and mosh kids also prefer afternoons.
- Most chances for having killer afternoon show is in snob pit as it has highest average energy scores for afternoon.
- Head singer is partly right about the full moon as although the full moon shows on average have highest crowd energy but difference between different moon phases is negligible.
- Best performing outfit is leather but difference is trivial.
- mosh pit is definitely outdoors as it suffer most in storms and rains and rainy overall do affect the energy of the crowd.
- The price sensitivity mentioned on pg.3 is definitely about the snob pit.
- Volume level is directly proportional to the crowd energy although each venue have different average volume levels but for all venues more volume translates to more energy.
- yup opener matters more than he think... drummer is right
- In general higher the price better the energy.
- yup snob audience wants overall experience but mosh audience do not give a shit about openers.
- Crowd Size do translates to more Crowd Energy in general and also for V_{Δ} .
- Saturday Shows are definitely better than rest for mosh pit
- Friday is the liveliest show for the monastery.
- For goths best day is sat and worst is Sunday.
- For Snob pit weekends are the best.

- Solving Contradictions
 1. Volume Matters more although more of both translates to more energy but volume affects more.
 2. In general Saturdays are the best but it also varies venue to venue.
 3. More Ticket price for V_Gamma is good but relatively bad for V_Beta and V_Delta and for V_Gamma there is hardly any difference
 4. Goths definitely becomes more lively at night but it is not true for all venues.
 5. Moon phase is real although its effect is trivial
 6. Rainy and stormy days are definitely worse than the rest.

Key findings for venues

- Monks absolute loud volume and low volume is an absolute bummer for them, despite this they have the lowest average volume level. Morning shows are better here. And Friday shows are better here.
- Goths definitely loves night shows especially on Sundays, but this venue in general had a lowest energy.
- Snob pit is the most exclusive venue as its ticket price is the highest and it even have more expensive VIP tickets. Here in the average volume level is the highest and shows here have highest energy as compared to other venues and it also have the highest crowd size in general. Here weekends afternoon are on next level. And audience here is more civilized and care for overall experience more.
- Mosh pit is a partly outdoors venue in general here crowd size is the lowest and here the ticket price is the cheapest as compared to all the venues and the its audience just wanna rage they literally can't care less about the complete experience.

Model Training

I chose the DecisionTreeRegression model for hyperparameter tuning because when I analyzed the relation between numerical features the linear correlations were extremely low. Whereas DecisionTreeRegression do not assumes linear relation and works great with non linear data by splitting data into segments. And the data contained mix of numerical and categorical features and decision trees can naturally handle this mix, also this model do not requires scaling of data unlike distance based algorithms. The hyperparameter were chosen by using Grid Search CV and comparing R2 and MSE scores of each combination.